Hemanth Kandimalla

Applied AI Scientist | Agent Alignment & Control

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EXPERIENCE

Celebal Technologies Sep 2026 - Present
Senior Software Engineer - Data Science | Applied AI Scientist

- Research agent alignment and control through context/persistent memory, workflow/dependency and memory/knowledge graphs; develop domain-specialized SLM/LLM adaptation, evaluation and deployment workflows.

Centre for Development of Advanced Computing (C-DAC) Aug 2023 - Aug 2026
Knowledge Associate | Applied AI Research

- Delivered 10 AI proofs of concept; adapted a 3B-parameter LLM and integrated retrieval/reranking, grounded generation, memory and safety in a FastAPI/vLLM application.
- Built state-machine agent harnesses with typed tool calls, verifier decisions, bounded repair, test/patch verification, permission gates and rollback.
- Measured serving TTFT/TPOT, throughput and GPU memory; evaluated prompt-injection, jailbreak and groundedness failures with red/blue agent loops.

MediaTek Jul 2022 - Jan 2023
Modem Software Engineer Intern

- Validated LTE/5G PHY-MAC behavior through protocol-log analysis and cross-layer hardware/software debugging.

SELECTED ENGINEERING

TrainScale Training systems / source

- Built PyTorch packing, LoRA/QLoRA and DDP/FSDP paths; restart checkpoints retain optimizer, scheduler and RNG state.
- Implemented Triton/autograd cross-entropy, RMSNorm and RoPE kernels with torch.compile and CUDA Graph execution paths.

SDKAtlas SDK analysis / source

- Built import-free AST analysis of signatures, inheritance and output-model evidence; reports unresolved cases and emits 14 CSV/JSONL/SQLite artifacts.

NeuralSkills Agent Skills / source

- Authored SDLC Agent Skills with typed JSON Schema records, evidence contracts, deterministic validation and behavioral evaluation specifications.

ENGINEERING FOCUS

Agent systems: Bounded harnesses, context/persistent memory, MCP/JSON-RPC and skill contracts
Model systems: PyTorch/Triton, autograd, torch.compile/CUDA Graphs, vLLM, LoRA and DDP/FSDP

SELECTED WRITING

Agentic State | Technical monograph on context, memory and agent control.

EDUCATION

Indian Institute of Technology Jammu
M.Tech, Communications and Signal Processing
CGPA: 8.34/10 2021 - 2023